

Task – 1: SMPS and UPS – Devices, Usage and Purpose

Device Type	Real-Life Function
SMPS Desktop Computer	Converts AC power into stable DC voltage required by computer components.
SMPS LED/LCD Monitor	Provides efficient and regulated DC power to the display circuit.
SMPS Wi-Fi Router	Converts household AC power into the low-voltage DC power required by the router.
UPS Desktop Computer	Provides temporary backup power during a power cut and prevents sudden shutdown.
UPS CCTV System	Keeps cameras and recording equipment running during power interruptions.

Task – 2: Working of SMPS and UPS

SMPS (Switched Mode Power Supply)

AC Input (230V) → SMPS → DC Output (12V / 5V / 3.3V) → Computer / Monitor

The SMPS takes standard AC power from the electrical supply and converts it into stable DC power required by the internal components of the system.

UPS (Uninterruptible Power Supply)

AC Input (230V) → UPS → Backup Power → Computer / CCTV Camera

The UPS provides backup power when there is a power failure and keeps connected devices operating for a short period.

Task – 3: Power Connectors – Types, Uses and Safety Tips

1. ATX 24-pin Motherboard Connector



- Provides the **main power supply to the motherboard**.
- Turn off and unplug the PC before connecting or disconnecting the connector.
- Match the clip with the correct position.
- Press the connector firmly until it locks properly.

- **Use:** Main power connection for the motherboard.

2. SATA Power Connector



- A **15-pin connector** used for supplying power to storage and optical drives.
- Used with **HDDs, SSDs, and DVD/CD drives**.
- Make sure the **L-shaped connector** is aligned correctly.
- Never force the connector into the socket.
- **Use:** Provides power to storage devices.

3. Molex Connector (4-pin)



- Commonly used with **older hard drives, optical drives, case fans, and other peripherals**.
- Hold the plastic connector instead of pulling the wires while disconnecting.
- Pull the connector straight out without applying excessive force.
- **Use:** Powers older devices and accessories.

4. PCIe 6/8-pin Connector



- Supplies additional power to **graphics cards (GPUs)**.
- Common types include **6-pin, 8-pin, and 6+2-pin connectors**.
- Use only the appropriate PCIe cable supplied by the PSU.

- Make sure the connector is completely inserted before switching on the PC.
- **Use:** Provides extra power for high-performance graphics cards.

Task – 4: Steps to Connect SMPS to a PC

1. Connect the Motherboard



Connect the **24-pin ATX power cable** from the SMPS to the motherboard's 24-pin power socket.

2. Connect CPU Power



Attach the **4+4-pin (8-pin) EPS/CPU power cable** to the CPU power socket located near the processor.

3. Connect the Graphics Card (GPU)



4.

Connect the required **PCIe 6-pin, 8-pin (6+2), or 12V-2x6/16-pin cable**, depending on the graphics card.

5. Connect Storage Devices



Use **SATA power connectors** from the SMPS for SATA SSDs and HDDs. A separate SATA data cable is used to connect each drive to the motherboard.

Note: M.2 NVMe SSDs do not require a separate SMPS power cable.